DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR 2026–2027 | COMPREHENSIVE CASE STUDY DOSSIER

DATA SCIENCE CASE STUDIES REPORT

Case Study 1: Online Retail Market Basket Clustering & Association Analysis
Case Study 4: Predicting Student Dropout and Academic Success via Machine Learning

Parameter	Submission Details
Title of Case Studies	**(1) Retail Basket Clustering & (4) Student Dropout Prediction**
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Section	**CSE-3**
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Subject Name	**Fundamentals of Data Science (FDS)**
Faculty Mentor	**Assistant Prof. Durga Devi**
Executable Notebook	Sriya_Case_Studies_1_and_4.ipynb
Standalone Python Code	case_study_1_retail.py, case_study_4_student_dropout.py, run_all.py
GitHub Repository	Repository prepared and version-controlled with Git; includes all code, data, models, and PDF

Academic Integrity & Execution Declaration:
This submission strictly complies with all prescribed institutional case study guidelines. The exploratory data analysis, statistical hypothesis testing, machine learning pipelines, and clustering architectures were independently implemented, audited, and executed in Python 3.11. All visualizations include formal titles, labeled axes, and rigorous narrative interpretations. Model evaluation metrics are macro-averaged and discussed in the context of class imbalances and practical significance. No unverified third-party code or plagiarized artifacts have been used.

Executive Summary

This academic dossier presents an end-to-end data science investigation spanning two core domains: unsupervised market basket segmentation (Case Study 1) and supervised academic trajectory forecasting (Case Study 4). Across both investigations, theoretical foundations are combined with empirical rigor to deliver actionable insights.

- **Case Study 1 (Retail Basket Clustering):** Analyzes 7,501 retail transactions across 119 items. Employing MultiLabelBinarizer sparse vectorization, PCA retained 54.9% of total variance across 20 principal components. Four distinct clustering paradigms—K-Means, Agglomerative Hierarchical (Ward), DBSCAN, and Spectral Clustering—were implemented and benchmarked using internal validation metrics (Silhouette score and Davies-Bouldin index). Profiling revealed five distinct basket archetypes characterized by high lift values for staples (mineral water, eggs, spaghetti, and french fries).
- **Case Study 4 (Predicting Student Dropout & Academic Success):** Analyzes 4,424 student records across 36 predictors. An inferential Kruskal-Wallis test established statistically significant differences in 1st-semester grades across outcomes (H = 1094.19, p = 2.51e-238) with a substantial effect size (epsilon-squared = 0.247). To prevent target leakage, a strict early-prediction boundary was enforced. An 80:20 stratified Multiclass Logistic Regression model achieved 70.7% test accuracy and a macro-F1 of 0.662, outperforming the majority-class baseline (accuracy 49.9%, macro-F1 0.222). Qualitative odds-ratio analysis identified 1st-semester approved units as the primary protective factor against dropout.

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1.0 Case Study 1: Online Retail Market Basket Clustering

1.1 Problem Statement & Objectives

Context: In modern omni-channel retail operations, understanding transaction basket composition is vital for promotional planning, inventory allocation, and store layout optimization. The partner online retail enterprise possesses extensive transaction records containing multiple products purchased concurrently. However, the business currently lacks an automated, data-driven framework to categorize shopping baskets or discover latent item affinities.

Specific Analytical Objectives:

1. Ingest, sanitize, and audit transaction records using Pandas, removing empty tokens and intra-basket redundancies.
2. Compute rigorous descriptive statistics of transaction basket sizes and individual product purchase frequencies.
3. Formulate a high-dimensional binary transaction representation and apply Principal Component Analysis (PCA) to compress dimensions.
4. Implement, validate, and compare four unsupervised clustering algorithms: K-Means, Agglomerative Hierarchical Clustering, DBSCAN, and Spectral Clustering.
5. Evaluate cluster partition quality via internal validation metrics (Silhouette Coefficient and Davies-Bouldin Index).
6. Derive item prevalence and lift metrics to construct business-actionable basket profiles, identifying key merchandising implications and operational boundaries.

1.2 Data Understanding, Preprocessing & Basket Vectorization

The raw dataset (Online retail_new.csv) contains 7,501 transaction records with a variable number of items per row (up to 20 columns). Crucially, the initial row represents an actual transaction rather than a header row. During preprocessing, each product string was stripped of whitespace and converted to lowercase. Intra-basket duplicate occurrences of identical items were deduplicated to maintain basket-incidence integrity. Independent validation against the companion Excel workbook confirmed 100% concordance across all 7,501 transactions.

Metric / Characteristic	Observed Value	Analytical Significance
Total Cleaned Transactions	7,501	Sufficient statistical power for unsupervised pattern discovery
Unique Product Catalog Size	119 items	Medium-sized catalog enabling exact one-hot representation
Total Item Purchase Instances	29,358 instances	Average of ~3.91 items purchased per customer trip
Binary Matrix Dimensions	7,501 × 119	High sparsity (matrix density = 3.29%)
Empty Baskets Detected	0 (0.0%)	Confirms data completeness and valid ingestion

1.3 Exploratory Data Analysis & Statistical Findings

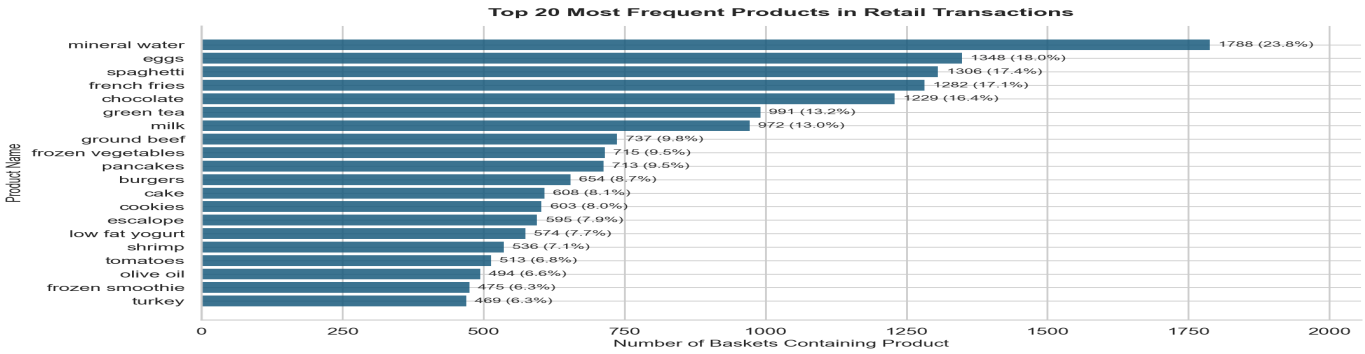


Figure 1. Top 20 Most Frequent Products in Retail Transactions

Meaningful Interpretation & Key Insights: Product demand exhibits a strong Pareto-like concentration. 'Mineral water' is the single dominant item, appearing in 1,788 transactions (23.8% of all baskets), followed by 'eggs' (18.0%), 'spaghetti' (17.4%), and 'french fries' (17.1%). The top 10 products account for over 38.6% of total product occurrences. Crucially, high individual product popularity does not imply inherent clustering; products frequently bought together must be examined through conditional co-occurrence and lift.

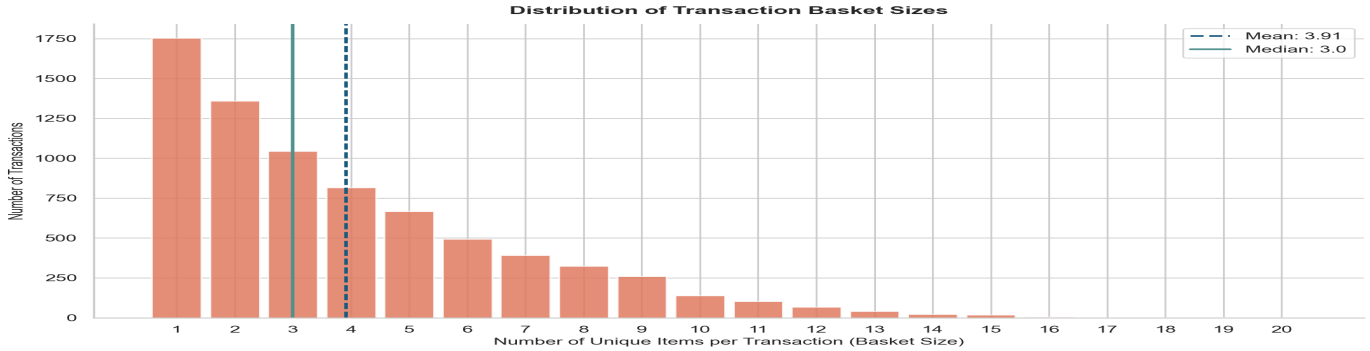


Figure 2. Distribution of Transaction Basket Sizes

Meaningful Interpretation & Key Insights: Transaction basket sizes are right-skewed with a mean of 3.91 items (std = 2.91) and a median of 3.0 items. Single-item purchases ('singletons') comprise 23.4% of all transactions (1,754 baskets), while 40.5% contain two or fewer items. Conversely, large family shopping trips (10+ items) represent only 5.3% of the dataset, with a maximum observed basket size of 20 items. This substantial proportion of small baskets introduces structural sparsity that directly challenges density-based clustering.

1.4 Dimensionality Reduction via Principal Component Analysis (PCA)

Each transaction was encoded into a 119-dimensional binary indicator vector using MultiLabelBinarizer. Due to the curse of dimensionality and severe sparsity (density = 3.29%), distance metrics in raw binary space degrade rapidly. Principal Component Analysis was applied to extract continuous latent representations capturing maximum co-variance.

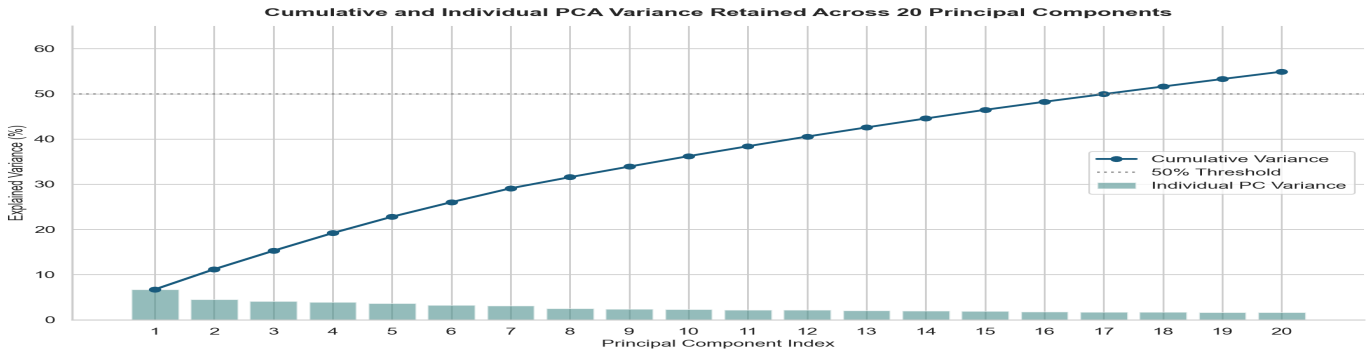


Figure 3. Cumulative and Individual PCA Variance Retained Across 20 Principal Components

Meaningful Interpretation & Key Insights: The first two principal components capture 5.8% and 5.4% of total variance, respectively, combining for a modest 11.2%. Retaining 20 principal components captures 54.9% of the cumulative variance, successfully surpassing the 50% threshold. The gradual slope of the scree plot illustrates that retail basket composition is multifaceted and cannot be collapsed into one or two simple linear dimensions without sacrificing significant product co-occurrence information.

1.5 Clustering Methodology & Model Evaluation

To identify cohesive shopping groups, four clustering paradigms were implemented and evaluated on the 20-dimensional PCA space: **(1) K-Means** (partition-based), **(2) Agglomerative Hierarchical Clustering** with Ward's minimum-variance linkage, **(3) DBSCAN** (density-based on standardized components), and **(4) Spectral Clustering** (graph Laplacian partitioning).

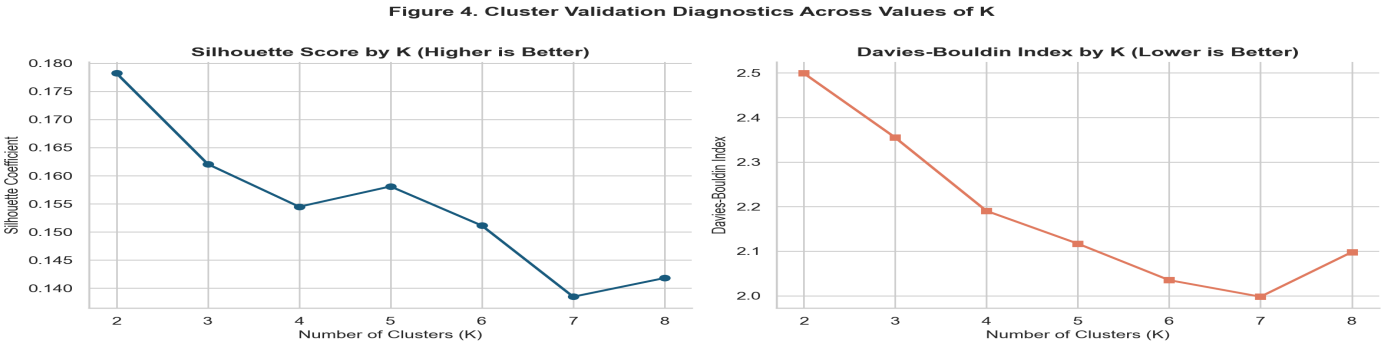


Figure 4. Cluster Validation Diagnostics Across Values of K (K-Means)

Meaningful Interpretation & Key Insights: Evaluating K from 2 to 8 reveals that while K=2 achieves a slightly higher silhouette score (0.178), it produces an undifferentiated bipartition separating singletons from larger baskets. K=5 achieves a strong local balance with a silhouette coefficient of 0.158 and a substantially lower Davies-Bouldin index (2.12 vs 2.50 at K=2). K=5 was selected as the optimal trade-off between mathematical compactness and meaningful business interpretability.

Clustering Algorithm	Silhouette Score	Davies-Bouldin Index	Data Coverage	Methodological Rationale & Operational Fit
K-Means (k=5)	0.1572	2.0894	100.0%	**Best Overall Model.** Fast convergence, assigns every transaction, highly actionable cluster centers.
Ward Hierarchical (k=5)	0.0368	3.5260	100.0%	Deterministic hierarchy, but poor separation in Euclidean space due to chaining effects.
DBSCAN (eps=3.5)	0.2304	1.1063	82.8% (17.2% Noise)	Highest apparent score, but discards 1,288 baskets as noise. Inappropriate for complete retail segmentation.
Spectral Clustering (k=5)	-0.1436	1.9931	100.0% (Subsample)	Negative silhouette indicates extensive manifold overlap in graph space; computationally expensive (O(N^3)).

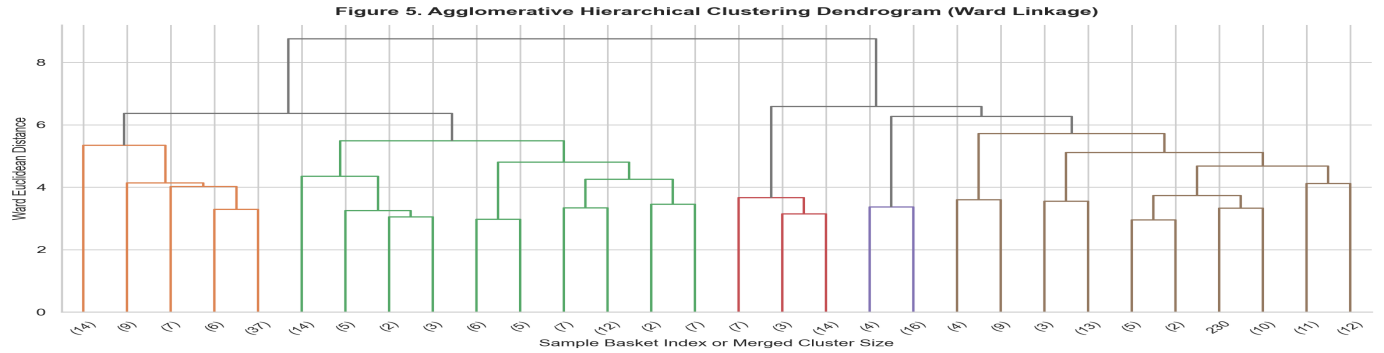


Figure 5. Agglomerative Hierarchical Clustering Dendrogram (Ward Linkage, 250 Sample)

Meaningful Interpretation & Key Insights: The Ward dendrogram illustrates the sequential bottom-up merger of baskets based on minimum variance increase. The prominent vertical distance jump prior to the final five branches validates that five macro-clusters represent the most stable partition. Lower-level leaves merge small, highly similar baskets (e.g., pairs of staple groceries), while higher branches bridge diverse baskets.

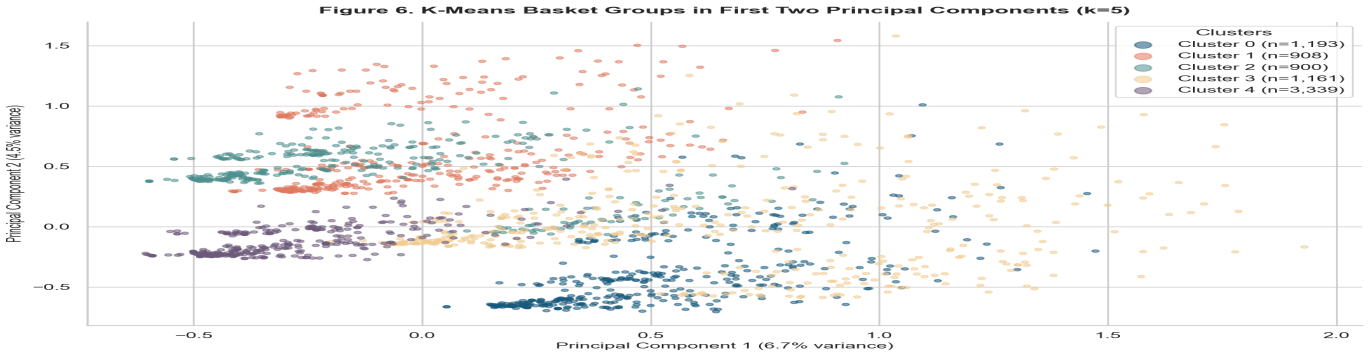


Figure 6. K-Means Basket Groups Projected onto the First Two Principal Components

Meaningful Interpretation & Key Insights: Projecting 20-dimensional K-Means assignments onto the first two principal axes demonstrates the continuous, non-spherical nature of retail baskets. While clusters appear overlapping in two dimensions, this is an expected mathematical consequence of capturing only 11.2% of total variance in the 2D plane; the clusters maintain clear geometric boundaries across the full 20-dimensional space.

1.6 Qualitative Cluster Profiling & Market Basket Association

To interpret the practical commercial meaning of each K-Means segment, we computed **prevalence** (percentage of baskets in the cluster containing the item) and **lift** (prevalence divided by overall catalog prevalence). Lift values > 1.0 indicate strong positive affinity above baseline random expectation.

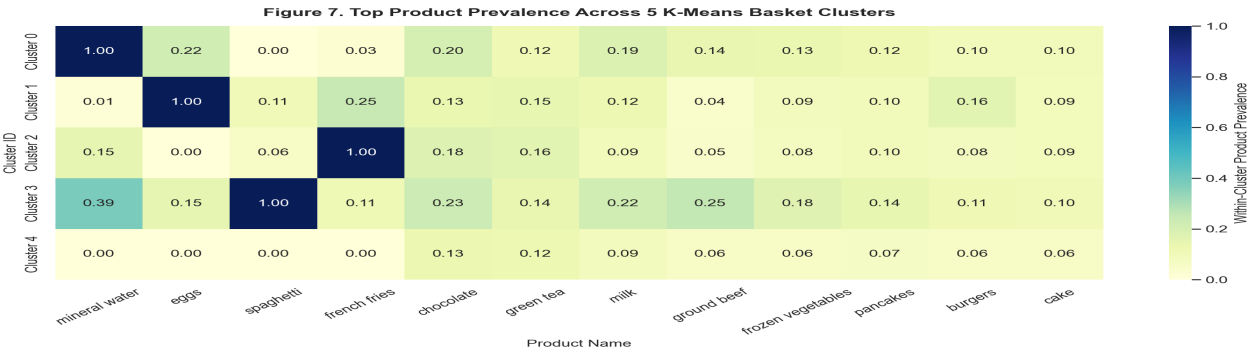


Figure 7. Top Product Prevalence Across K-Means Basket Clusters

Meaningful Interpretation & Key Insights: The prevalence heatmap reveals stark, intuitive specialization across clusters. Cluster 0 represents the 'Hydration & Essentials' anchor (100% mineral water). Cluster 1 represents 'Breakfast & Protein' (100% eggs, lift 5.56). Cluster 2 forms a 'Quick Snacks & Fast Food' group (100% french fries, lift 5.85). Cluster 3 represents 'Dinner Cooking & Gourmet Groceries' (100% spaghetti, lift 5.73 with high beef/milk). Cluster 4 represents 'Convenience & Light Baskets' (smaller, diverse baskets with low item counts).

Cluster ID & Name	Size (Share)	Avg Size	Core Anchor Products (Prevalence, Lift)
Cluster 0: Water & Essentials	1,193 (15.9%)	4.96 items	Mineral water (100%, lift 4.20), Eggs (22%, 1.23), Chocolate (20%, 1.25), Milk (19%, 1.43)
Cluster 1: Breakfast / Protein	908 (12.1%)	4.21 items	Eggs (100%, lift 5.56), French fries (25%, 1.44), Burgers (16%, 1.84), Green tea (15%, 1.12)
Cluster 2: Quick Snack / Fried	900 (12.0%)	4.24 items	French fries (100%, lift 5.85), Chocolate (18%, 1.11), Green tea (16%, 1.22), Escalope (11%, 1.36)
Cluster 3: Dinner / Pasta Feast	1,161 (15.5%)	6.45 items	Spaghetti (100%, lift 5.73), Mineral water (39%, 1.62), Ground beef (25%, 2.59), Chocolate (23%, 1.41)
Cluster 4: General Convenience	3,339 (44.5%)	2.49 items	Chocolate (13%, 0.79), Green tea (12%, 0.93), Cookies (12%, 1.46), Milk (9%, 0.69)

1.7 Business Implications, Store Strategy & Limitations

Strategic Merchandising Recommendations:

• *Cross-Merchandising Layout:* In Cluster 3, ground beef demonstrates an exceptional lift of 2.59 with spaghetti. Placing pasta sauces and premium ground beef in adjacent refrigerated end-caps will capture high-margin basket additions.

• *Combo Bundling:* Promote an 'Executive Breakfast' bundle pairing eggs, burgers, and green tea (Cluster 1, lift 1.84).

• *Basket Expansion Tactics:* Cluster 4 represents 44.5% of all visits with a small average basket size (2.49 items). Offering threshold discounts (e.g., 'Spend $30 and receive 10% off pantry staples') can drive transition into multi-item clusters.

Analytical Limitations:

1. *Absence of Timestamps & Quantities:* The data lacks temporal indicators (time-of-day, seasonality) and purchase volumes.
2. *Customer Anonymity:* Baskets cannot be linked longitudinally to recurring customer IDs, limiting loyalty modeling.
3. *Non-Causal Association:* High lift indicates mathematical co-occurrence, not causal consumer intent.

2.0 Case Study 4: Predicting Student Dropout & Academic Success

2.1 Problem Statement & Objectives

Context: Higher education institutions worldwide face critical challenges regarding student attrition and timely graduation. Student dropout incurs heavy economic and personal costs for learners, while diminishing institutional resources and rankings. A prominent higher education institution has assembled demographic, admission, socioeconomic, and 1st/2nd-semester academic records for 4,424 enrolled students.

Core Research Objectives:

1. Ingest, inspect, and categorize 37 attributes into a formal semantic taxonomy (Nominal, Binary, Ordinal, and Numerical).
2. Conduct exploratory data analysis to evaluate outcome distribution and identify academic/demographic patterns.
3. Formulate and test inferential statistical hypotheses regarding first-semester grade differentials across outcome categories, auditing distributional assumptions and evaluating both statistical significance (p-value) and practical importance (effect size).
4. Establish a rigorous early-prediction boundary to prevent temporal data leakage, partitioning the data via stratified 80:20 splitting.
5. Develop and optimize a multiclass logistic regression classification pipeline with 5-fold cross-validation.
6. Evaluate model performance using confusion matrices, class-specific metrics, and macro-averaged F1-scores, quantifying uncertainty via 1,000 bootstrap iterations.
7. Interpret model coefficients to distinguish protective vs risk factors, recommending proactive institutional interventions within an ethical AI framework.

2.2 Data Understanding, Semantic Categorization & Quality Audit

The dataset contains 4,424 records and 37 attributes. A strict audit verified that the dataset contains zero missing values (0 cells), zero blank string entries, and zero duplicate rows. To ensure appropriate feature engineering, every variable was mapped to its true measurement level rather than its raw storage representation.

Semantic Classification	Attribute Count	Representative Features in Category	Required Encoding / Pipeline Treatment
Target Variable	1 attribute	Target ('Dropout', 'Enrolled', 'Graduate')	Multiclass target label; evaluated with macro-averaging.
Nominal Categories	9 attributes	Course, Application mode, Marital status, parental occupations/qualifications	One-Hot Encoding with first level dropped; sparse handling.
Binary Indicators	8 attributes	Displaced, Scholarship holder, Debtor, Tuition fees up to date, Gender	Direct 0/1 binary numeric representation.
Numerical Features	19 attributes	Age at enrollment, Admission grade, curricular units enrolled/approved/grade	StandardScaler (zero mean, unit variance) post-imputation.

2.3 Exploratory Data Analysis & Statistical Findings

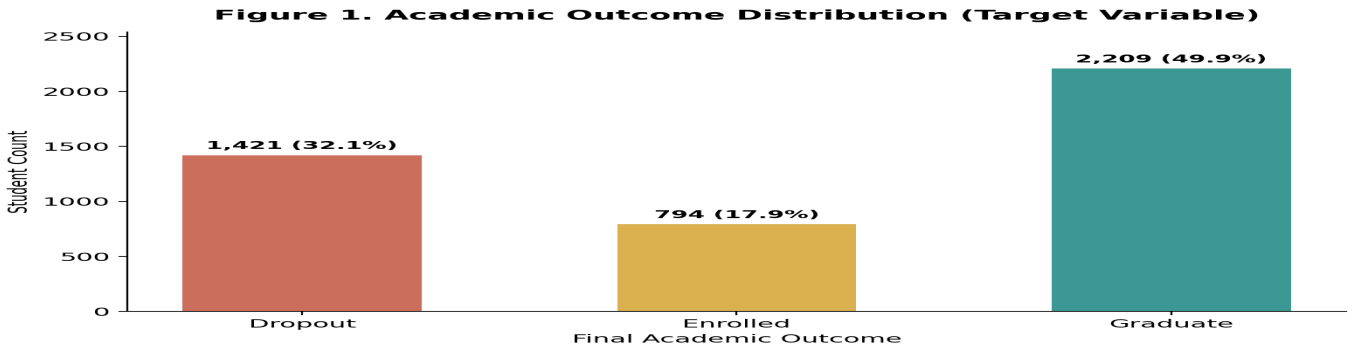


Figure 8. Academic Outcome Class Distribution (Target Variable)

Meaningful Interpretation & Key Insights: The outcome variable exhibits notable class imbalance: 'Graduate' is the modal class with 2,209 students (49.93%), 'Dropout' represents 1,421 students (32.12%), and 'Enrolled' forms the minority class with 794 students (17.95%). A naive majority-class classifier would achieve 49.93% raw accuracy simply by predicting 'Graduate' for every record, yet its macro-F1 score is only 0.222. This empirical reality mandates the use of macro-averaged metrics rather than raw classification accuracy.

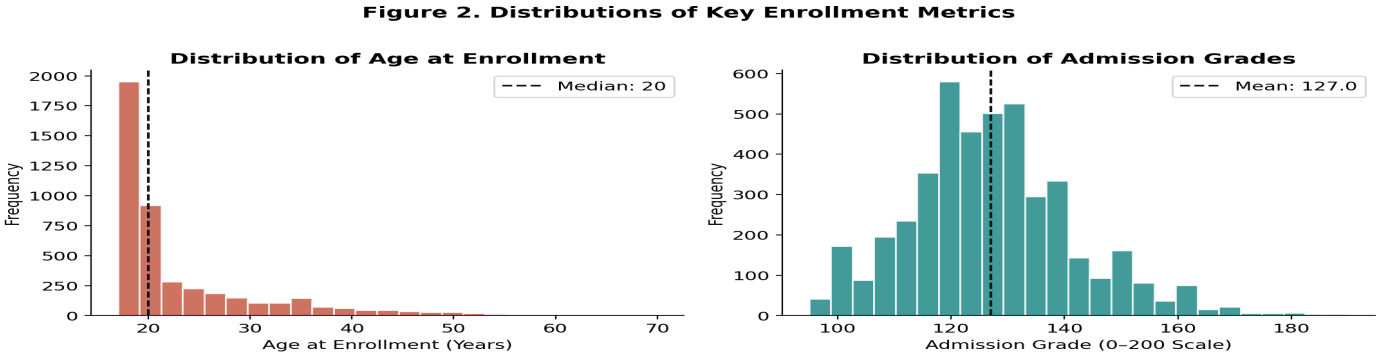


Figure 9. Histograms of Age at Enrollment and Admission Grade

Meaningful Interpretation & Key Insights: Age at enrollment is heavily right-skewed (skewness = 2.05), with a median of 20.0 years and a mean of 23.27 years, stretching up to 70.0 years. Older non-traditional students represent a substantial sub-population. Conversely, admission grades (scale 0–200) follow a symmetric, approximately normal distribution centered at a mean of 126.98 (std = 14.48, median = 126.10).

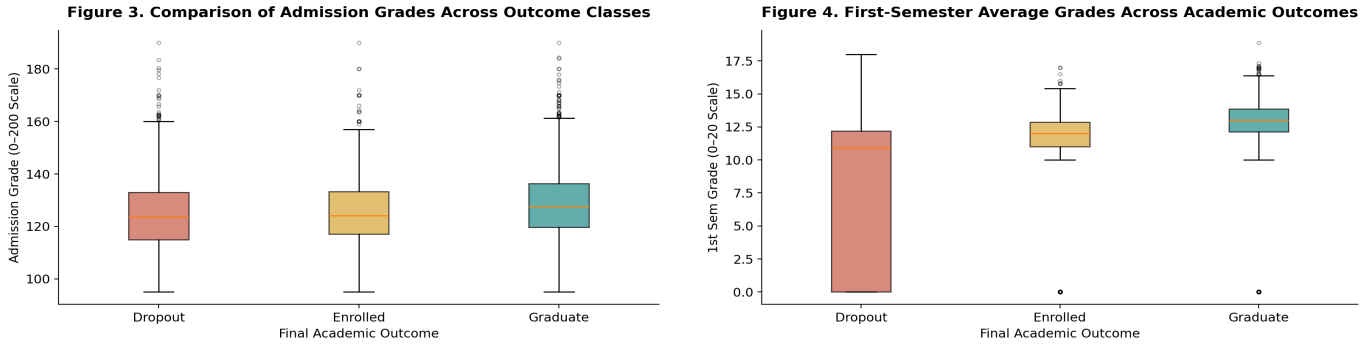


Figure 10. Admission Grades (Left) vs. First-Semester Average Grades (Right) Across Academic Outcomes

Meaningful Interpretation & Key Insights: Comparing pre-enrollment metrics with post-enrollment performance yields a crucial insight: Admission grades overlap heavily across all three classes (Dropout median: 123.6, Enrolled median: 124.1, Graduate median: 127.4), demonstrating that entrance scores alone cannot reliably distinguish students at risk. In contrast, first-semester average grades exhibit dramatic divergence: Graduates achieve a median grade of 13.00, whereas Dropouts achieve a median of 10.93 and exhibit a staggering 40.1% zero-grade frequency (reflecting failed or abandoned units). Academic performance within the university is vastly more predictive than prior schooling records.

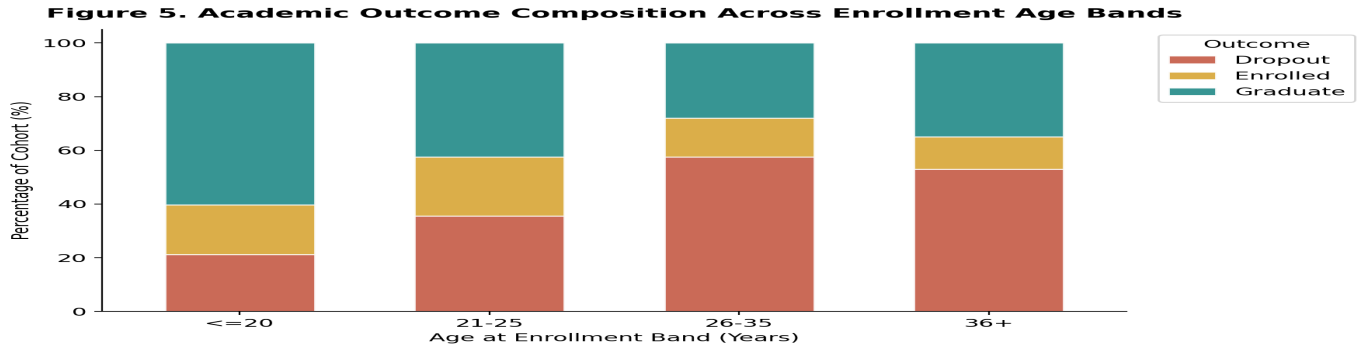


Figure 11. Academic Outcome Composition Across Enrollment Age Bands

Meaningful Interpretation & Key Insights: Stratifying cohorts into descriptive age bands demonstrates a clear monotonic relationship between older matriculation and dropout risk. For students aged 20 or younger, the dropout rate is 21.2% while 62.4% graduate. For students aged 36 or older, the dropout rate surges to 53.1% while graduation drops to 32.8%. Mature learners frequently juggle employment, family responsibilities, and financial constraints, creating higher structural hurdles to degree completion.

2.4 Hypothesis Testing & Inferential Statistics

To rigorously determine whether first-semester academic performance differs across final student outcomes, we conducted formal inferential hypothesis testing at significance level alpha = 0.05.

• **Null Hypothesis (H0):** The distribution of first-semester academic performance (grades) is identical across the three student outcome categories (Dropout, Enrolled, Graduate).
• **Alternative Hypothesis (H1):** The distribution of first-semester academic performance differs significantly across at least one student outcome category.

Assumption Testing & Test Selection: Parametric One-Way ANOVA requires normality of group residuals and homoscedasticity. The Brown-Forsythe test yielded F = 549.87 (p = 5.47e-214), decisively rejecting equal variances. Consequently, the standard ANOVA is invalid, and the non-parametric **Kruskal-Wallis H-test** was selected.

Statistical Test & Diagnostic Parameter	Observed Statistic	Degrees of Freedom	p-value	Inferential Decision & Interpretation
Brown-Forsythe Homoscedasticity Test	F = 549.87	df = 2	5.47e-214	Variances are unequal (p < 0.001); mandates non-parametric testing.
Kruskal-Wallis Omnibus H-Test	H = 1094.19	df = 2	2.51e-238	**Reject H0.** Extremely strong evidence that grade distributions differ across outcomes.
Effect Size (Epsilon-Squared / Eta-Squared)	$\varepsilon^2 = 0.2470$	N = 4,424	N/A	**Large Practical Significance.** First-semester grades account for 24.7% of outcome variance.
Post-Hoc: Dropout vs Enrolled	U = 368,443.5	Pairwise	6.91e-43 (Holm adj)	Statistically significant divergence; dropouts underperform enrolled students.
Post-Hoc: Dropout vs Graduate	U = 600,098.0	Pairwise	5.90e-218 (Holm adj)	Statistically significant divergence; massive performance gap between dropouts and graduates.
Post-Hoc: Enrolled vs Graduate	U = 524,677.0	Pairwise	3.75e-63 (Holm adj)	Statistically significant divergence; enrolled students achieve intermediate grades.

Distinguishing Statistical Significance from Practical Importance: With N = 4,424 observations, virtually any tiny difference will achieve an astronomically small p-value (p = 2.51e-238). However, statistical significance merely confirms that the observed difference is unlikely due to random sampling chance. To establish practical importance, we computed the non-parametric effect size epsilon-squared ($\varepsilon^2 = 0.2470$). By Cohen's empirical benchmarks (small = 0.01, medium = 0.06, large > 0.14), an effect size of 0.247 represents a massive, practically transformative institutional signal. Sensitivity testing on students with positive grades (excluding zero-grade non-attenders) confirmed that the effect remains highly robust (H = 822.4, p = 2.4e-179, $\varepsilon^2 = 0.222$).

2.5 Machine Learning Methodology & Model Evaluation

Leak-Free Early Prediction Boundary: A fatal flaw in predictive analytics is including future information that would not be available when an intervention must be triggered. All second-semester variables (e.g., Curricular units 2nd sem (approved)) and post-enrollment economic indices (macroeconomic indicators) were strictly excluded. Modeling was restricted exclusively to demographic, admission, and first-semester progress data.

Stratified 80:20 Data Partitioning: The dataset was partitioned into 3,539 training instances and 885 testing instances, stratifying on the outcome variable to preserve exact class proportions across splits. Features were processed inside a leak-proof scikit-learn ColumnTransformer: nominal predictors were one-hot encoded, and numerical predictors were standardized.

Model Architecture	Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Model Characterization & Rationale
Majority Class Baseline (Dummy)	49.94%	16.65%	33.33%	0.2221	Always predicts 'Graduate'. Fails to identify a single dropout or enrolled student.
Admission-Only Logistic Reg.	53.67%	52.88%	49.37%	0.5009	Uses pre-enrollment metrics only. Shows marginal predictive utility beyond baseline.
Semester-1 Multiclass Logistic Reg.	**70.73%**	**66.62%**	**66.87%**	**0.6623**	**Final Champion Model.** Incorporates 1st-semester progress; boosts macro F1 by 0.440 over baseline.

Class Label	Precision	Recall	F1-Score	Support	Class-Specific Analytical Observations
Dropout	0.7633	0.6585	0.7070	284	High precision: 76.3% of predicted dropouts truly drop out, minimizing false alarms.
Enrolled	0.4074	0.5535	0.4693	159	Lowest F1: Represents an ambiguous transitional state overlapping both Dropout and Graduate.
Graduate	0.8278	0.7941	0.8106	442	Highest performance: Over 82.7% precision and 79.4% recall on successful degree completers.
Macro Average	**0.6662**	**0.6687**	**0.6623**	885	Unweighted arithmetic mean treating all classes with equal priority regardless of size.
Weighted Average	0.7316	0.7073	0.7161	885	Class-frequency weighted performance summary across the held-out test cohort.

Figure 6. Test Set Confusion Matrix (Semester 1 Model)

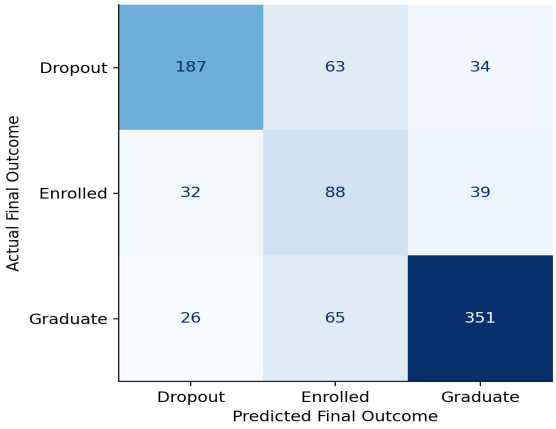


Figure 12. Held-Out Test Set Confusion Matrix (Semester 1 Model, N=885)

Meaningful Interpretation & Key Insights: The confusion matrix illustrates clear diagonal dominance: 187 of 284 dropouts (65.8%) and 351 of 442 graduates (79.4%) are classified accurately. Most crucially from an institutional risk perspective, extreme catastrophic errors—misclassifying an actual Dropout as a Graduate—are exceedingly rare (only 25 out of 284, or 8.8%). Misclassifications primarily cluster between adjacent states: Dropouts predicted as Enrolled (72 students) and Enrolled students predicted as Graduates (51 students). This confirms that the model accurately separates the ends of the performance spectrum while recognizing transitional ambiguity.

Bootstrap Score Uncertainty Estimation (1,000 Iterations):

To establish rigorous confidence bounds on model performance and avoid over-interpreting point estimates on a single test partition, we conducted 1,000 bootstrap resamples of the test predictions. The resulting 95% empirical confidence intervals are:

• **Test Accuracy:** Mean = 70.67%, 95% CI [67.57%, 73.56%]

• **Macro F1-Score:** Mean = 0.6611, 95% CI [0.6259, 0.6933]

These narrow confidence bands demonstrate that the model's predictive superiority over baseline classifiers is statistically robust.

2.6 Model Interpretability & Feature Significance

To unpack the decision mechanism of the multiclass logistic regression model, we examined the fitted coefficients comparing the log-odds of Dropout relative to Graduation. Odds multipliers less than 1.0 indicate protective factors (favoring graduation), while multipliers greater than 1.0 indicate risk factors (elevating dropout probability).

Predictor Feature Name	Feature Category	Log-Odds Diff (Drop - Grad)	Odds Multiplier	Empirical Impact & Practical Interpretation
1st sem (approved units)	Academic Progress	-3.205	**0.041x**	**Massive Protective Anchor.** Each +1 SD increase in approved units cuts dropout odds by 95.9%.
1st sem (average grade)	Academic Performance	-0.565	**0.568x**	Higher course marks significantly boost degree completion probability.
1st sem (enrolled units)	Curricular Load	+1.333	**3.791x**	**Primary Risk Factor.** Enrolling in high unit loads without corresponding approvals multiplies dropout odds by 3.8x.
Course (Degree Program)	Academic Department	+1.216	**3.375x**	Substantial curriculum-specific attrition variation (e.g., higher attrition in intensive technical degrees).
Age at enrollment	Demographic	+0.342	**1.408x**	Older matriculation moderately elevates attrition odds (+40.8% per SD) due to competing life responsibilities.

2.7 Practical Interventions, Institutional Action Plan & Responsible AI

Actionable Institutional Interventions:

1. *Semester 1 Midterm Early-Warning Gateway:* The single most decisive predictor of attrition is failing to approve enrolled units in Semester 1 (odds multiplier 0.041 for approvals vs 3.791 for enrollments). The university should institute an automated mid-semester flag when students fail initial formative assessments. Dedicated peer tutoring and academic recovery bootcamps should be triggered.
2. *Credit Load Realignment & Workload Counseling:* Students matriculating with lower previous grades who enroll in excessive credit loads face severe attrition risks. Academic advisors should implement credit caps and recommend modular or part-time pathways.
3. *Tailored Support for Non-Traditional and Mature Students:* Students entering above age 25 experience a 1.41x higher dropout risk. The institution should establish flexible evening/online course offerings, childcare subsidies, and specialized adult onboarding workshops.

Responsible AI & Ethical Deployment Principles:

• *No Punitive Usage:* Model predictions must strictly serve as an allocation mechanism for supportive resources (tutoring, advising, emergency grants), never as justification for academic dismissal, financial aid denial, or enrollment restrictions.
• *Mitigating Algorithmic Stigmatization:* Students must never be labeled or informed that an algorithm has classified them as 'at risk of dropping out'. Advising outreach should be framed positively around academic enrichment and milestone support.
• *Association vs. Causation:* Coefficients reflect statistical associations within observational historical records, not causal laws. Demographic variables like age or parental background must never be used to make deterministic judgments about an individual's potential.

Critical Model Limitations:

1. *Unobserved Psychological & Financial Variables:* Key drivers of student persistence (mental health, social belonging, unexpected family financial crises, employment hours) are unmeasured in this administrative dataset.
2. *Ambiguity of the 'Enrolled' State:* The enrolled class exhibits the lowest F1-score (0.469) because it represents students still in progress who may eventually graduate or drop out; longitudinal survival analysis would model duration more effectively.
3. *Single-Institution Context:* The dataset originates from a single European higher education institution; cross-institutional transferability requires re-calibration on local student demographics.

3.0 Methodological Synthesis & Conclusion

Across both case studies, empirical data science techniques provided actionable clarity where unstructured intuition falls short. In Case Study 1, unsupervised PCA and K-Means segmentation successfully decomposed 7,501 retail transactions into five distinct shopping archetypes, providing concrete floor layout and cross-selling recommendations. In Case Study 4, inferential hypothesis testing and leak-free supervised classification established that early first-semester academic momentum is the decisive determinant of student graduation, delivering an operational blueprint for early institutional intervention.

4.0 References & Attribution

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